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Incident Timeline and Evidence Collection for Small Teams

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Problem Description

Cybersecurity investigations often require examining logs collected from different systems. Differences in log formats, timestamps, and system clock settings make it difficult to reconstruct the exact sequence of events and investigate incidents efficiently.

Issues to be addressed:

- Logs from different sources are stored in different formats.
- Inconsistent timestamps make event correlation difficult.
- Evidence integrity must be maintained during collection and analysis.
- Manual log analysis is time-consuming.
- Malformed or unsupported log entries may be overlooked.

Background

System and application logs provide valuable information during cybersecurity incident investigations. However, reconstructing an incident requires combining records from multiple sources while maintaining evidence integrity and accounting for differences in timestamps. A structured approach can help investigators understand the sequence of events more efficiently.

Key Requirements:

- Ingest and parse at least two log formats.
- Preserve original logs and generate SHA-256 hashes.
- Normalize timestamps to UTC while retaining original values.
- Identify timestamp inconsistencies and parsing gaps.
- Build a searchable chronological incident timeline.
- Generate an incident report for further investigation.

Literature & RFC Review

Literature	Key Contributions	Research Gaps
NIST SP 800-86	Guidelines for digital forensics and incident response.	Does not prescribe a ready-made lightweight tool for this project.
NIST SP 800-92	Best practices for security log management.	Focuses on overall log management rather than our specific workflow.
RFC 5424	Defines a standard syslog message format.	Does not provide cross-source incident timeline analysis.
RFC 3339	Standardizes Internet date and time representations.	A timestamp format alone cannot resolve incorrect system clocks.
Plaso	Generates timelines from digital forensic data.	Our project will investigate a simpler, focused workflow for two log formats.
Timesketch	Supports collaborative forensic timeline analysis.	Our project will focus on an integrated workflow for small-team triage.

Proposed Solution

Vigilance

Incident Timeline and Evidence Collection

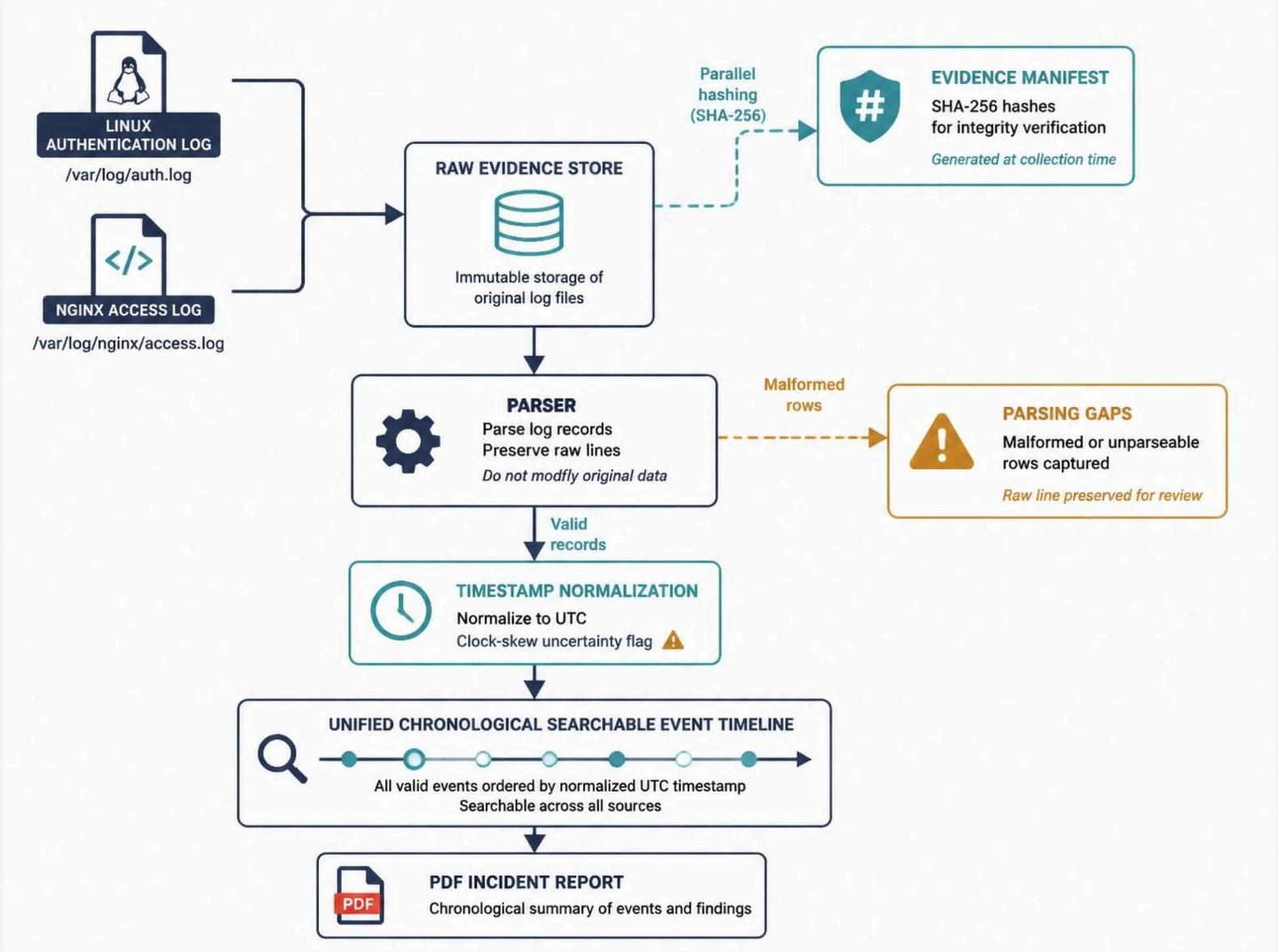
Vigilance is a lightweight tool that brings logs from different sources into one searchable timeline. It preserves original evidence, standardizes timestamps, highlights missing or uncertain information, and generates an incident report.

What?	Why?	How?
Scattered Logs	Make incident investigation difficult.	Combine records from two log formats.
Evidence Integrity	Evidence must remain verifiable.	Preserve originals and generate SHA-256 hashes.
Inconsistent Timestamps	Can lead to incorrect event sequences.	Normalize timestamps to UTC and flag uncertainty.
Parsing Gaps	Important events may be overlooked.	Identify and report unreadable or unsupported records.
Manual Investigation	Takes time and effort.	Provide a searchable timeline and exportable report.

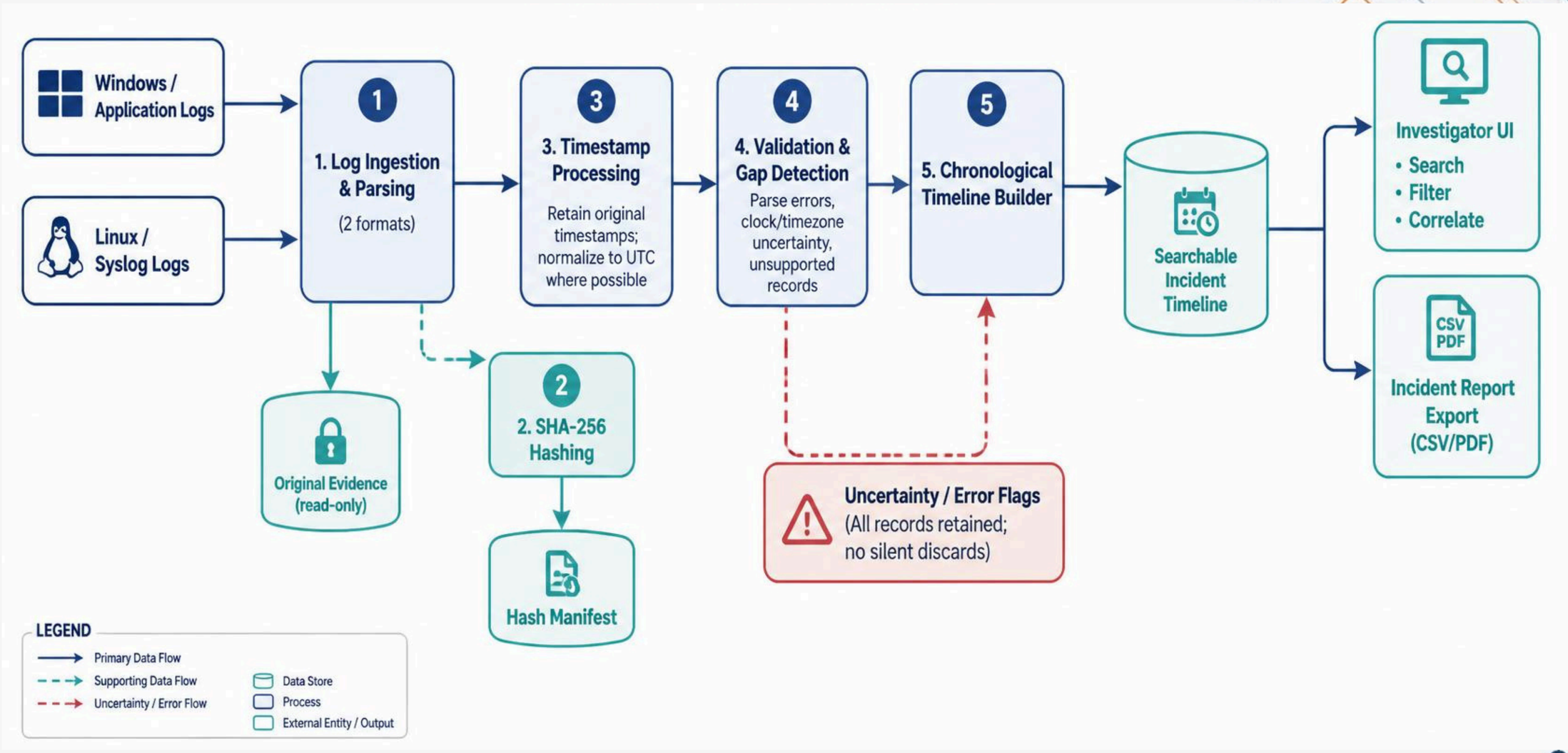
Optimization Proposed

Existing Challenge	Vigilance's Approach
Logs stored separately	Combine events in one timeline
Different time zones	Convert timestamps to UTC where possible
Evidence may be altered	Preserve originals and verify hashes
Some records cannot be read	Flag parsing errors and gaps
Manual investigation takes time	Provide search, filters and reports

Solution Architecture & Design



Data Flow Diagram



Deployment Timeline

Phase	Activity
1	Requirement analysis and design
2	Log collection and parsing
3	Evidence hashing and timestamp processing
4	Timeline interface and report generation
5	Testing with three incident scenarios
6	Evaluation, demonstration and documentation

Bibliography

[1] NIST SP 800-86: Guide to Integrating Forensic Techniques into Incident Response

[2] NIST SP 800-92: Guide to Computer Security Log Management

[3] RFC 5424: The Syslog Protocol

[4] RFC 3339: Date and Time on the Internet

[5] Plaso: Digital Forensic Timeline Tool

[6] Timesketch: Collaborative Timeline Analysis

Thank You for Your Attention!